

Developing Energy-Efficient Diesel Solutions Using Semiconductors

If the automotive industry has to achieve and improve the efficiency and emission norms, electronics has to play a major role, which it is well equipped to do so now

With the cost of fuel increasing day by day, there is a rise in the requirement of diesel vehicles, especially passenger cars. Earlier, pollution and vehicle maintenance was a major issue, but now this has been taken care of by the improved engine technology.

The set targets for fuel efficiency cannot be achieved without innovating and improving all the different components that contribute to the fuel or energy consumption. Meant for energy generation, the engine contributes a lot to inefficiency due to various factors; the role of engine accessories, transmission, aerodynamics, etc cannot be discounted as well.

Along with the improvements to the power-train, efficiency improvements to the other consumers of energy are also in focus, without which it is very difficult to meet the performance requirements of the Euro-6 or similar standards. Electronics can and will play a major role in supporting this drive towards higher efficiency with its ability to sense accurately, process data fast enough and control the components with precision.

Starting from the sensors such as tyre-pressure monitoring systems to the nitrogen-oxide (NOx) detector inside the exhaust system, there are hundreds of components that enable a system to measure and report dynamic parameters that are required for processing. Pedal position, throttle position, manifold air pressure and barometric pressure are some

of the parameters that are normally measured. Since the efficiency of this control system very much depends on the accuracy of most of these input parameters, these sensors should be reliable and durable enough.

Sensor technology has improved

and the newer set of sensor families such as integrated giant magneto-resistive (iGMR) sensors provide enhanced measurement precision. However, it is not just enough to convert parameters



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such as pressure, density, position and angle into electric current or voltage. Some of the sensors have the capability to send these parameters wirelessly or through standard serial peripheral interface, etc. The microcontrollers in the engine control units (ECUs) have in-built interfaces to receive the data from standard protocols. These microcontrollers also have a set of sophisticated analogue-to-digital converters (ADCs) that are fine-tuned for the automotive power-train-specific application.

To achieve the norms of Euro-6 and beyond, the new technologies and innovations are supported by sensors that detect the parameters such as NOx density and ammonia density in the case of new diesel engines. These outputs constitute a number of variable parameters that should be fed into the ECU, and the data bandwidth and data

processing requirements are not less for any standard. The microcontrollers are also improving, with chipsets going multi-core with hardware accelerations implemented on-chip to achieve higher performance requirements.

As the measurement precision of

sensors is important, the precision and accuracy of the output controls can also influence the efficiency. Since the reduction in carbon-dioxide emission directly influences the fuel-to-energy conversion efficiency positively, the ability to precisely inject the fuel during the various stages of combustion cycle is provided by the microcontrollers.

Today, microcontrollers are equipped with timer arrays that are capable of offloading most timing-critical tasks of complex trigger generations based on the inputs such as shaft positions.

Even more important are the power controllers, insulated gate bipolar transistors (IGBTs) and SPI driver for enhanced relay (SPIDER) control that finally control the glow-plugs, solenoids and piezo injectors. The innovations in these chipset families are contributing to faster response time, durability and power efficiency.

It is important to understand that optimal use of electronics will provide a best fit to the users as well as the manufacturers in terms of cost and feature ratio. Finally, with more and more gadgets and features adding every day, there is a huge scope to increase the semiconductor content per car. ●



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